Balanced Diet and Nutrition

Introduction to Nutrition

Nutrition is the science of how the body uses food to grow, work, repair itself and stay healthy. A balanced diet is a diet that provides all the essential nutrients in the right proportions, along with enough water and fibre, to meet the body's needs for energy, growth and protection from disease. Nutrients are the useful substances in food. Good nutrition improves immunity, learning ability, energy levels and mood, while poor nutrition leads to malnutrition, which includes undernutrition, such as stunting and anaemia, and overnutrition, such as obesity and diabetes. India faces a double burden of malnutrition, with both undernutrition and rising obesity. National Nutrition Week is observed from 1 to 7 September, and September is Poshan Maah.

The Main Nutrients and Food Groups

Food provides six classes of nutrients: carbohydrates, proteins, fats, vitamins, minerals and water, with dietary fibre also important. Based on their function, foods are grouped into energy-giving foods (carbohydrates and fats), body-building foods (proteins) and protective foods (vitamins and minerals).

Food is also grouped into major food groups. These include cereals, millets and whole grains such as rice, wheat, ragi, jowar and bajra, which provide carbohydrates; pulses, legumes and nuts such as dals, chickpeas and groundnuts, which provide protein; milk and milk products such as curd, buttermilk and paneer, which give calcium and protein; eggs, fish and meat, rich in protein; vegetables and fruits, which provide vitamins, minerals and fibre; and fats and oils, such as groundnut oil, sesame oil, ghee and coconut, which provide energy and essential fatty acids. The ICMR-National Institute of Nutrition in Hyderabad published the revised Dietary Guidelines for Indians in 2024, with 17 guidelines and the "My Plate for the Day" model, recommending a diet largely based on a variety of foods, more vegetables, fruits and whole grains, and less sugar, salt and oil.

Carbohydrates, Proteins and Fats

Carbohydrates are the body's main source of energy; each gram gives 4 kilocalories. They are found in rice, wheat, millets, potatoes and sugar. Complex carbohydrates in whole grains digest slowly and are better than refined sugar. Proteins give 4 kilocalories per gram. Fats give 9 kilocalories per gram, the most concentrated energy, and help absorb vitamins A, D, E and K. Healthy fats come from nuts, seeds, fish and vegetable oils, while trans fats in some fried and packaged foods should be avoided.

Why Does the Body Need Protein?

The body needs protein for growth and repair of tissues. Proteins are long chains of amino acids; there are 20 amino acids, of which nine are essential, meaning that the body cannot make them and they must come from food. Protein builds and repairs muscles, skin, hair and nails; makes enzymes that run chemical reactions; forms hormones such as insulin; makes antibodies that fight infection; carries oxygen as haemoglobin; and can supply energy when needed. A typical adult needs about 0.8 to 1 gram of protein per kilogram of body weight each day, more for growing children, pregnant women and athletes. Rich sources include dals and pulses, soya, milk, curd, paneer, eggs, fish, chicken, nuts and sprouts. Vegetarians can combine cereals and pulses, such as rice and sambar, or idli with dal, to get all essential amino acids. Lack of protein causes a deficiency disease called kwashiorkor in children.

Vitamins and Minerals

Vitamins are needed in small amounts. Vitamin A protects eyesight, and its lack causes night blindness. The B vitamins help release energy and keep the nerves healthy. Vitamin C, found in amla, guava and citrus fruits, helps wounds heal, and its lack causes scurvy. Vitamin D, made in the skin by sunlight, helps absorb calcium, and its lack causes rickets. Vitamin K helps blood clotting. Minerals include calcium for bones and teeth, iron for haemoglobin (lack causes anaemia), iodine for the thyroid (lack causes goitre, which is why India promotes iodised salt), zinc and potassium.

Two Foods Rich in Fibre

Dietary fibre is the part of plant food that the body cannot digest. It prevents constipation, helps control blood sugar and cholesterol and gives a feeling of fullness. Two foods rich in fibre are whole grains such as ragi, oats and whole wheat, and pulses and legumes such as rajma, chana and dals. Others include vegetables like beans, carrots and spinach, fruits such as guava, apple and papaya with skin, nuts and seeds. Adults should have about 25 to 30 grams of fibre a day. The World Health Organization advises eating at least 400 grams of fruits and vegetables a day.

Water and Healthy Habits

Water makes up much of the body and helps digestion and temperature control. A balanced plate should be colourful, include some protein at every meal, limit sugary drinks, junk food and excess salt, and include regular meal times. Reading food labels helps in choosing healthy products.

Nutrition Programmes in India and Tamil Nadu

India's POSHAN Abhiyaan, launched in 2018, aims to reduce stunting, undernutrition and anaemia. The PM POSHAN scheme provides mid-day meals in schools. The mid-day meal was first introduced in Tamil Nadu in the 1950s under Chief Minister K. Kamaraj and was expanded as the Noon Meal Scheme in 1982, and the state launched a Chief Minister's Breakfast Scheme for primary school children in 2022. Traditional Tamil meals served on a banana leaf, such as rice, sambar, rasam, kootu and curd, are examples of a balanced diet.

Challenges and Future

Challenges include junk food, lack of awareness, poverty and food adulteration. The future of nutrition involves personalised diets, fortified foods and promotion of millets.

Quick Facts

- The main nutrients are carbohydrates, proteins, fats, vitamins, minerals and water.
- Protein is needed for growth, repair of tissues, enzymes, hormones and antibodies.
- Fibre-rich foods include whole grains such as ragi and oats, pulses, fruits and vegetables.
- Carbohydrates and proteins give 4 kcal per gram and fats give 9 kcal per gram.
- ICMR-NIN published the Dietary Guidelines for Indians in 2024.